High-throughput screening using Scipion-chem

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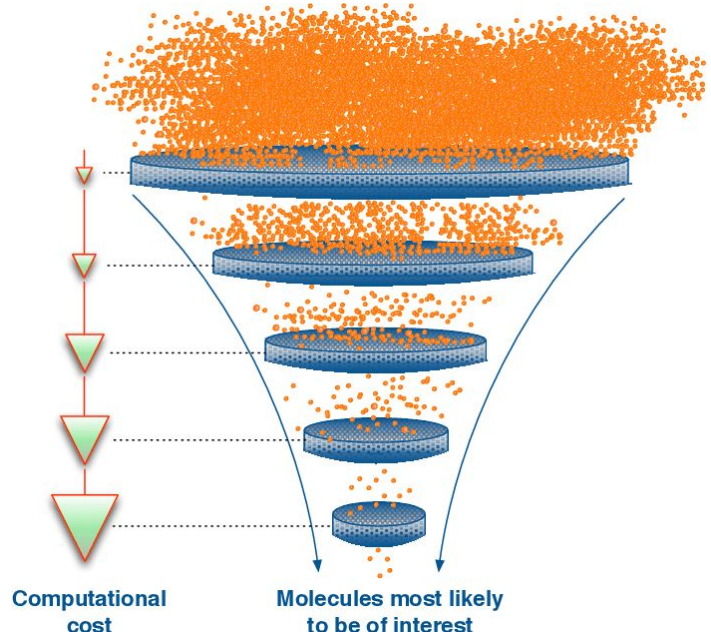


High-throughput virtual screening

- Screens **very large chemical libraries**, potentially containing millions or billions of compounds
- **Goal:** Rapidly prioritize promising candidates for further computational or experimental evaluation

- **Why use machine learning?**
  - Docking millions of compounds can be **computationally expensive**
  - ML models **learn representations** of proteins and molecules to **estimate compatibility**
  - Enable **large-scale screening** before a more detailed docking or experimental validation
  - **Reduce search space** for computationally intensive methods

*Key idea: ML-based screening **prioritizes candidates efficiently**, but predictions require **further validation***

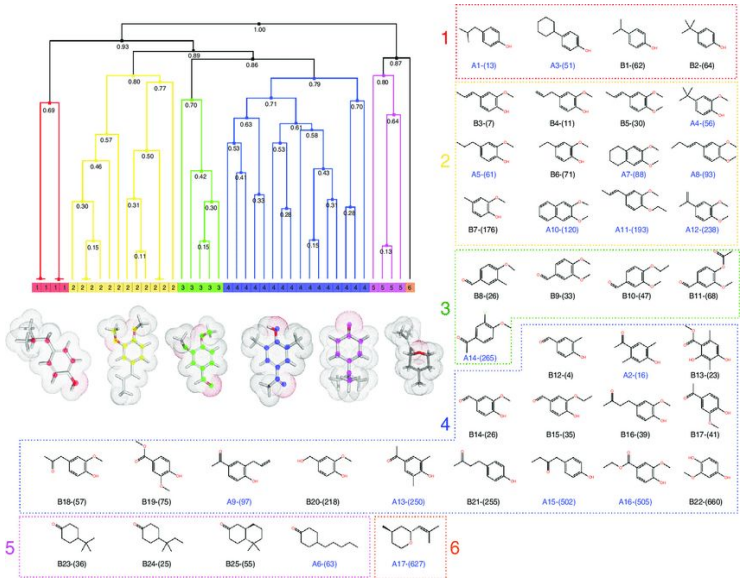


Molecule clustering: pre-filtering step

- **Goal:** Reduce redundancy in large chemical libraries before screening

- **Molecular fingerprints:** Encode chemical features of molecules to enable similarity comparisons
- **Clustering:** Group molecules according to their chemical similarity
- **Representative selection:** Selects representative compounds from each cluster, reducing library size while retaining chemical diversity

- **Example: BitBIRCH**
  - An efficient clustering algorithm designed to handle large datasets
  - Uses molecular fingerprints to organize compounds into similarity-based clusters
  - Helps reduce the number of molecules that need to be screened individually

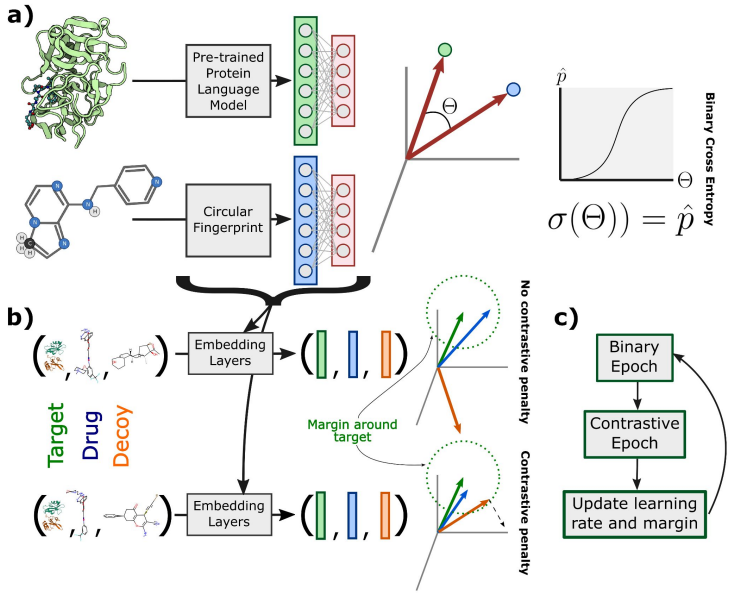


*Key idea: Clustering **reduces the library size** by grouping similar molecules; it does **not predict their biological activity***

Sequence-based models

- **Inputs:** Protein amino acid sequence + ligand representation
- Use learned representations to estimate compatibility
- **Fast inference** makes it suitable for large-scale screening
- Do not require 3D protein structure as input

- **Example: ConPLex**
  - Uses **contrastive learning** to model protein-ligand compatibility
  - **Inputs:** sequence and SMILES
  - Learns a shared embedding space in which compatible protein–ligand pairs are brought closer together
  - Enables rapid prioritization of candidate compounds across protein targets



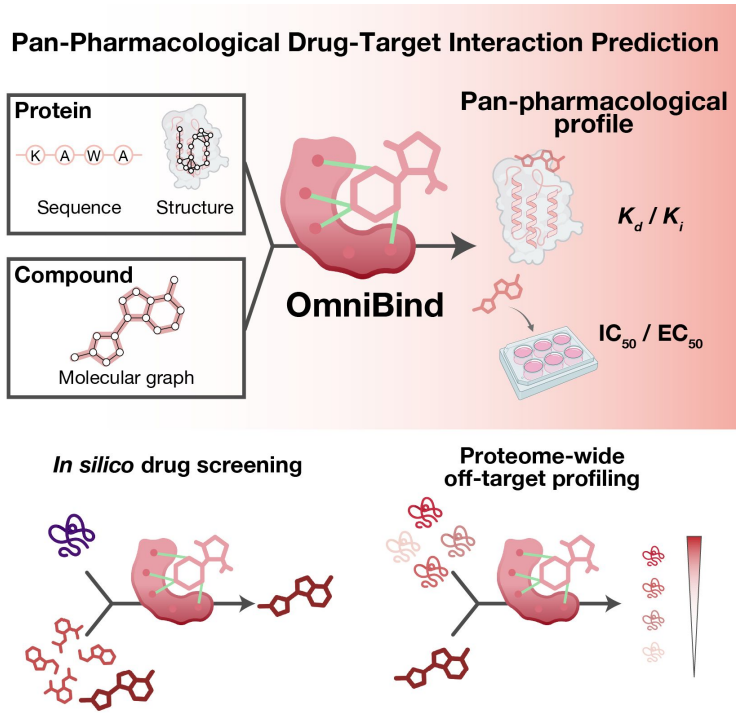
*<u>Key idea</u>: Sequence-based models can screen compounds **without requiring** an experimentally determined or predicted **protein structure***

Structure-based models

- **Inputs:** Protein 3D structure + ligand representation
- Incorporate **structural information** about the protein and ligand
- Estimate compatibility based on features of the binding site and molecule
- Can take advantage of information that is not available from the protein sequence alone

- **Example: OmniBind**
  - Structure-aware approach to predict protein–ligand compatibility
  - **Inputs:** protein structure and ligand molecular graph
  - Learns representations of protein–ligand interactions from structural information
  - Can be used to prioritize candidate compounds

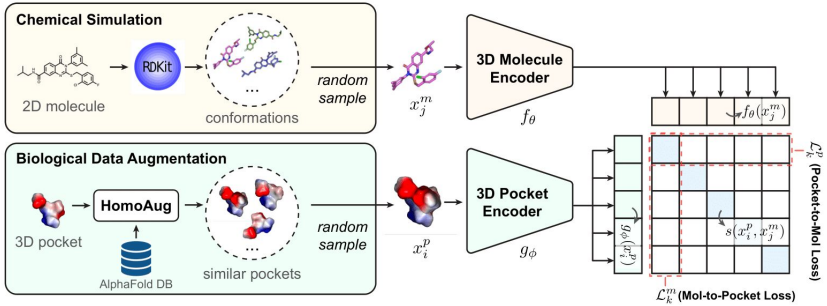
*Key idea: Structure-based models incorporate **3D information**, but their predictions can **depend** on the **quality** and conformation of the input structure*



Contrastive retrieval models

- Learn representations of protein-binding pockets and ligands in a **shared embedding space**
- Candidate ligands are ranked according to their similarity to the representation of a target binding pocket
- This approach enables efficient screening of very large chemical libraries

- **Example: DrugCLIP**
  - Uses **contrastive learning** for protein-ligand matching
  - **Inputs:** protein binding pocket and ligand representation
  - Learns to align representations of compatible protein-ligand pairs
  - Formulates virtual screening as a **large-scale retrieval and ranking problem**



<u>Key idea</u>: Instead of explicitly docking every molecule, retrieval models rapidly ***identify*** *ligands whose* ***learned representations match a target binding pocket***

Comparison of ML-based screening approaches

	Sequence-based	Structure-based	Contrastive retrieval
Example tool	**ConPLex**	**OmniBind**	**DrugCLIP**
Protein input	Amino acid sequence	3D protein structure	Binding pocket
Ligand input	SMILES	Molecular graph	Molecular representation
Main approach	Learns protein and ligand representations	Incorporates protein structural info	Matches protein pockets and ligands in a shared embedding space
Main advantage	No structure	Incorporates 3D information	Efficient large-scale similarity search
Main limitation	Does not explicitly account for 3D binding-site geometry	Depends on structural quality and the selected protein conformation	Retrieval scores do not directly measure binding affinity